# AWS VPC PEERING

Private Connectivity Between Two Amazon Virtual Private Clouds

EWB Cloud Labs  Student Project Report

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**AWS Region:** Asia Pacific (Mumbai)  ap-south-1

**Project Status:** Implemented and connectivity tested

This report documents the VPCs, subnets, route tables, VPC peering connection, bastion access, and ICMP connectivity tests shown in the supplied screenshots.

# 1. Project Overview

**Objective.** Create two isolated Amazon VPC networks and connect them using VPC Peering so instances can communicate through private IP addresses. Use a bastion host as the SSH entry point to reach private instances.

**Why this matters.** This pattern can support communication between application environments or shared internal services while keeping the application instances without public IP addresses. Routing and security-group rules still control whether traffic is permitted.

**Region:** ap-south-1 (Mumbai).

**Peering connection:** ewb-a-b-peer — **ID:** pcx-0ccbf4d4e42ed055b — **Status:** Active.

# 2. Architecture and Resource Inventory

| Resource | Name / CIDR / Address | Purpose |
|---|---|---|
| VPC A | ewb-vpc-a — 10.10.0.0/16 | Network for bastion and App-A |
| VPC B | ewb-vpc-b — 10.20.0.0/16 | Network for App-B |
| Public subnet | ewb-a-public — 10.10.1.0/24 | Hosts the bastion |
| Private subnet A | ewb-a-private — 10.10.2.0/24 | Hosts App-A |
| Private subnet B | ewb-b-private — 10.20.1.0/24 | Hosts App-B |
| Bastion | ewb-bastion — 10.10.1.10 (public IP enabled) | SSH jump host |
| App-A | ewb-app-a — 10.10.2.10 (private IP) | Private EC2 instance in VPC A |
| App-B | ewb-app-b — 10.20.1.10 (private IP) | Private EC2 instance in VPC B |

## Architecture diagram

| | | |
|---|---|---|
| **VPC A — 10.10.0.0/16**<br>Public subnet: 10.10.1.0/24<br>Bastion: 10.10.1.10 | **VPC Peering**<br>ewb-a-b-peer<br>pcx-0ccbf4d4e42ed055b | **VPC B — 10.20.0.0/16**<br>Private subnet: 10.20.1.0/24<br>App-B: 10.20.1.10 |
| **Private subnet A**<br>10.10.2.0/24<br>App-A: 10.10.2.10 | Private traffic uses routes in both directions. | **Connectivity result**<br>App-A ↔ App-B<br>ICMP successful |

Note: The diagram is a simplified logical view; the screenshots in later sections are the evidence for the actual AWS resources.

# 3. VPC and Subnet Configuration

The screenshots show both lab VPCs in the Available state. VPC A contains a public subnet and a private subnet; VPC B contains the private subnet used by App-B.

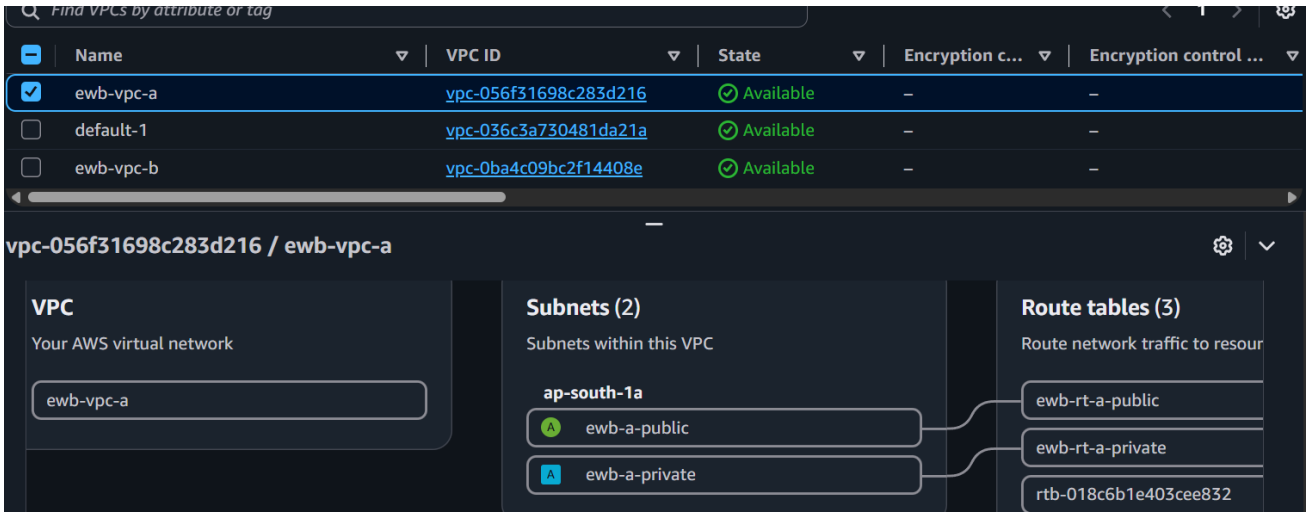



*Figure 1. VPC A selected in the AWS VPC console, showing its subnet and route-table layout.*

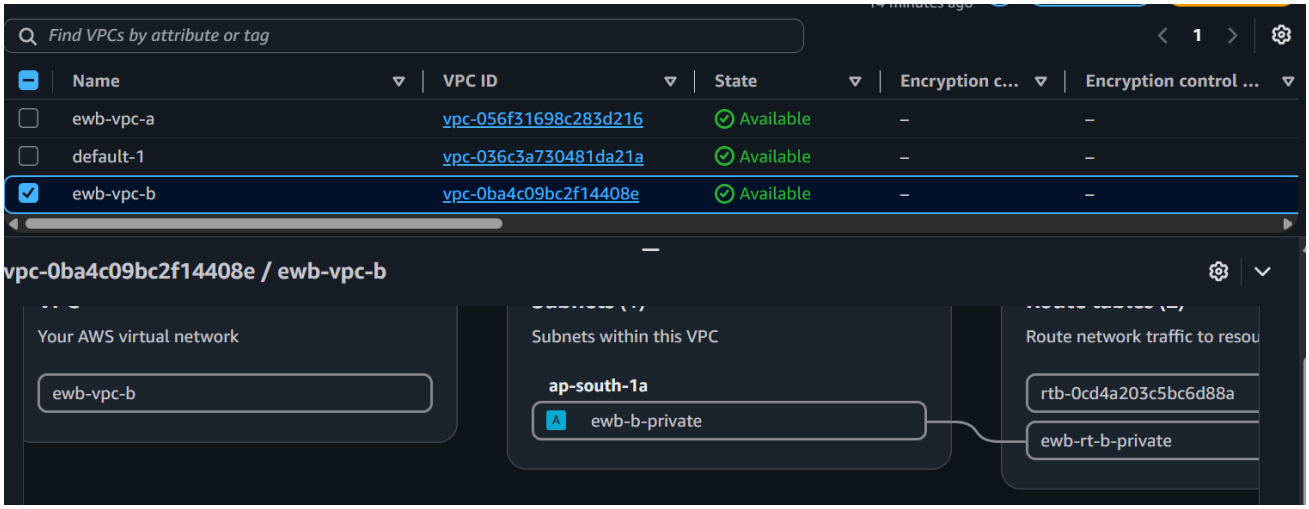



*Figure 2. VPC B selected in the AWS VPC console, showing ewb-b-private and its route table.*

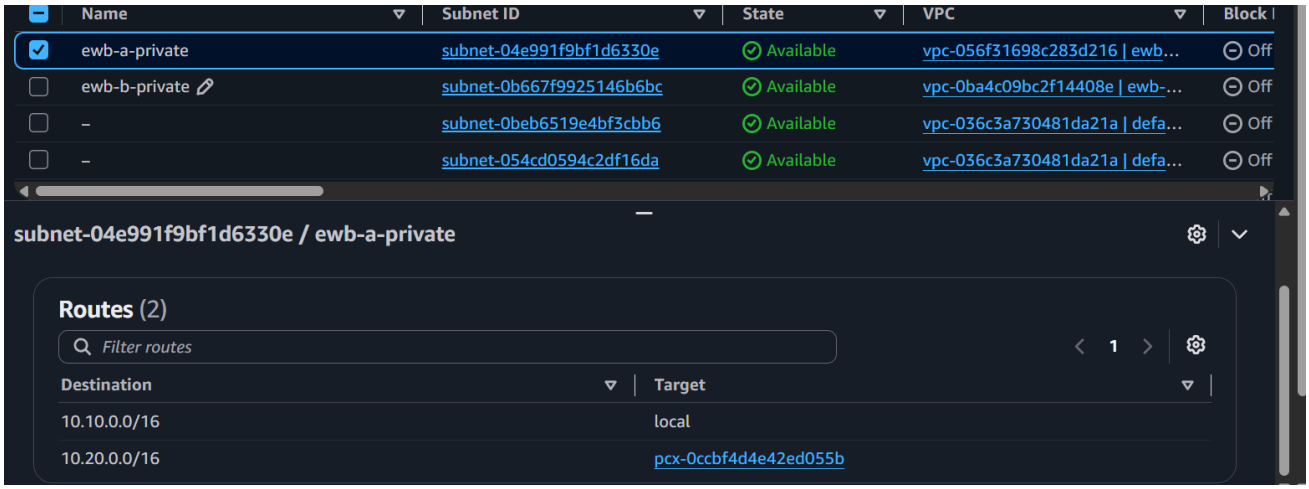



*Figure 3. Route information for ewb-a-private, including the route to VPC B through peering.*

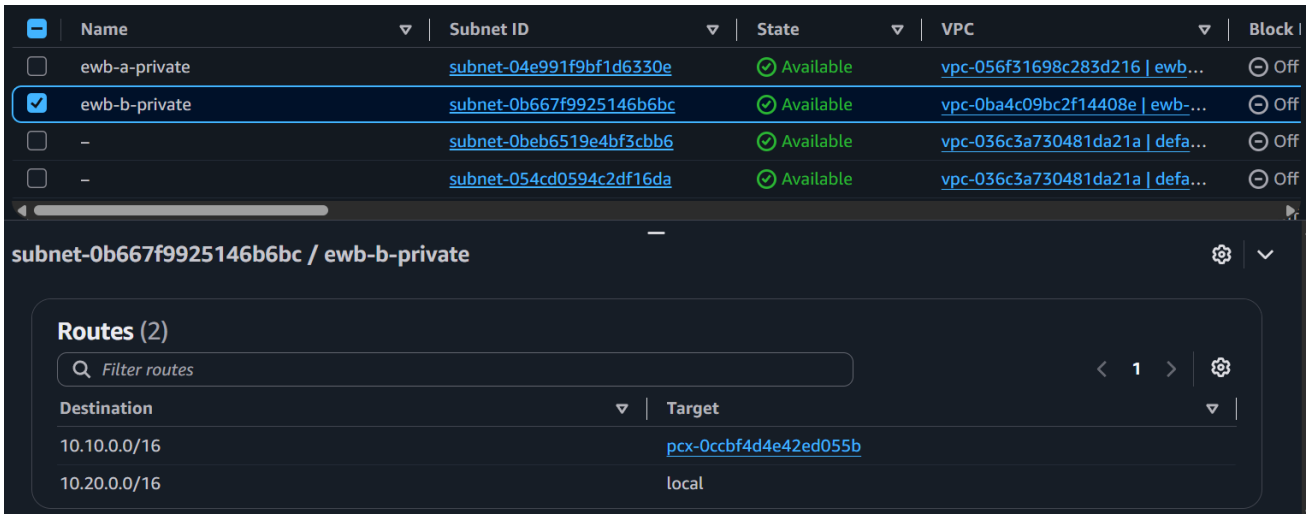



*Figure 4. Route information for ewb-b-private, including the route to VPC A through peering.*

# 4. VPC Peering Connection and Route Tables

The peering connection ewb-a-b-peer is shown as Active. Its connection ID is pcx-0ccbf4d4e42ed055b.

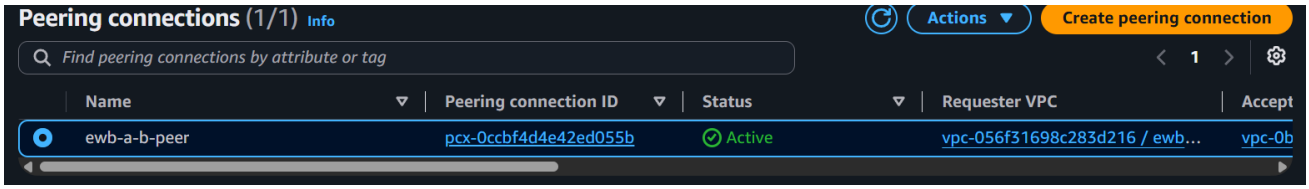



*Figure 5. VPC Peering connections page showing ewb-a-b-peer in Active status.*

The following routes were configured to provide private routing between the CIDR ranges:

| Route table | Destination | Target |
|---|---|---|
| ewb-rt-a-private | 10.20.0.0/16 | pcx-0ccbf4d4e42ed055b |
| ewb-rt-b-private | 10.10.0.0/16 | pcx-0ccbf4d4e42ed055b |
| ewb-rt-a-public | 10.20.0.0/16 | pcx-0ccbf4d4e42ed055b |

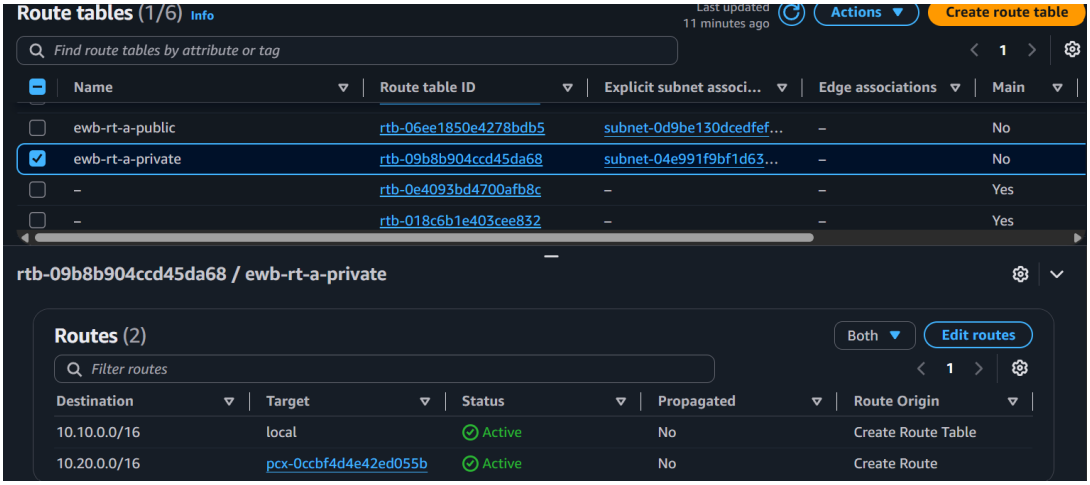



*Figure 6. ewb-rt-a-private: local route and peering route to 10.20.0.0/16.*

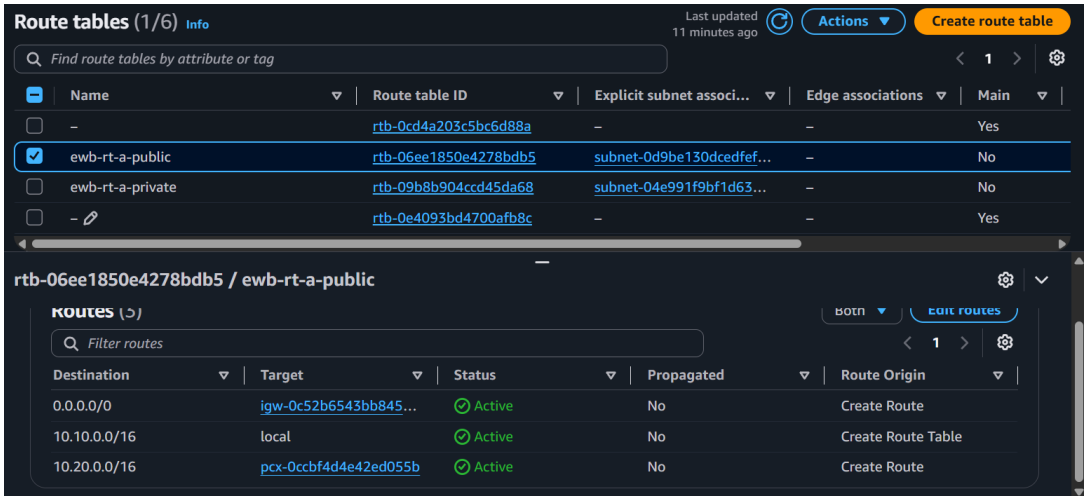



*Figure 7. ewb-rt-a-public: local route, internet gateway default route, and peering route.*

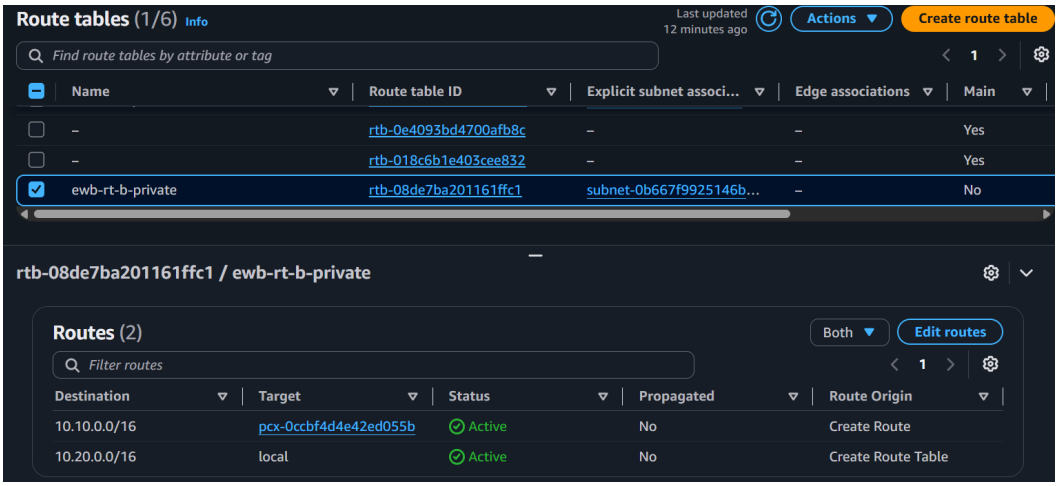



*Figure 8. ewb-rt-b-private: local route and peering route to 10.10.0.0/16.*

# 5. Bastion Host and SSH Access

The terminal screenshot shows a successful SSH session to ewb-bastion running Amazon Linux 2023. The student used a WSL SSH configuration with ProxyJump to access the private instances through the bastion.




*Figure 9. Successful SSH connection to ewb-bastion; the prompt shows private address 10.10.1.10.*

# 6. Connectivity Testing

The baseline test from App-A to App-B failed before the peering routes were in place. After the peering connection and routes were configured, the same ping test succeeded, and the reverse-direction ping also succeeded.

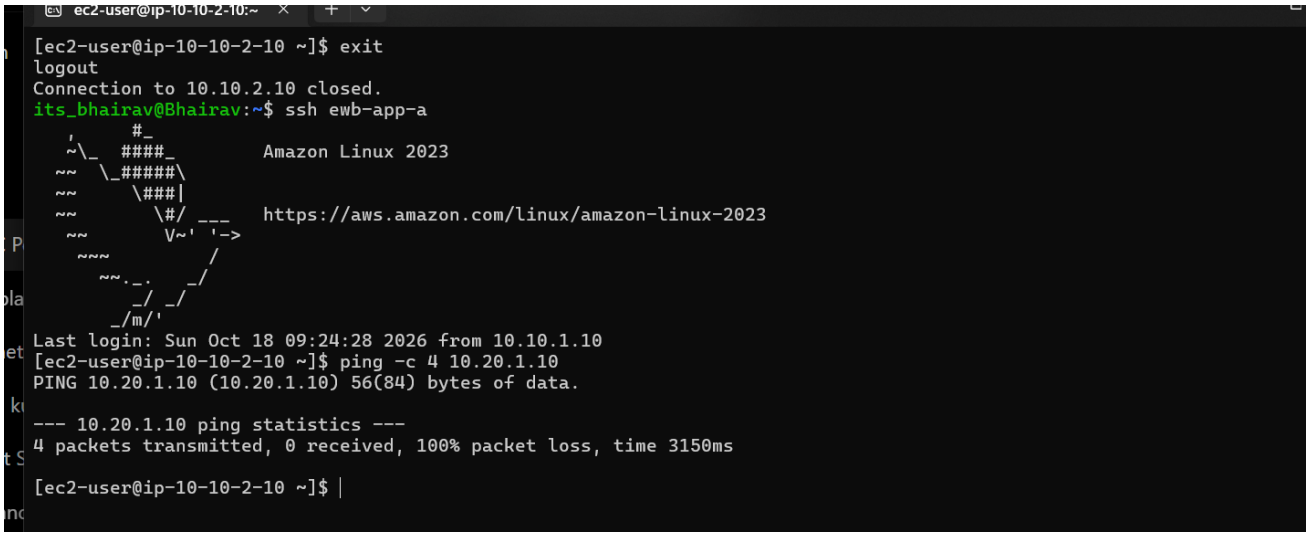



*Figure 10. Baseline test from App-A to 10.20.1.10: four packets sent, zero received, 100% packet loss.*

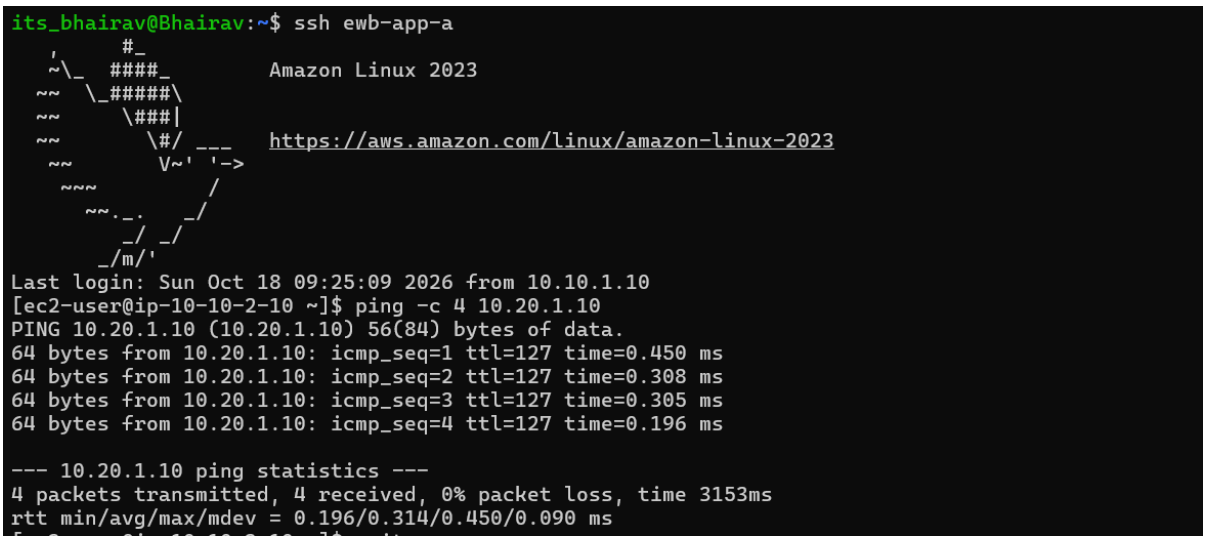



*Figure 11. App-A to App-B after configuration: four packets received, 0% packet loss.*

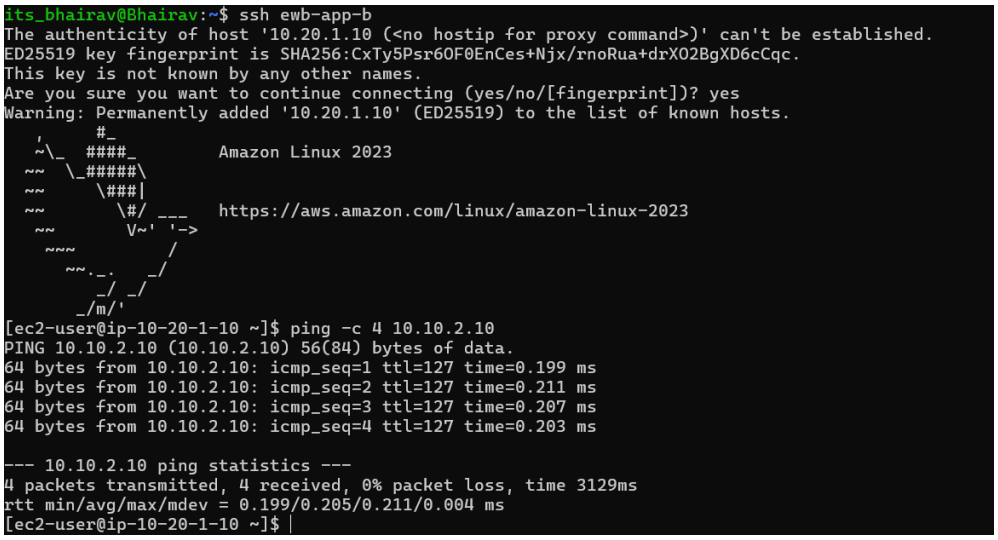



*Figure 12. App-B to App-A after configuration: four packets received, 0% packet loss.*

# 7. Results and Lessons Learned

**Result.** The supplied screenshots demonstrate an Active VPC peering connection, the three required peering routes, successful SSH access to the bastion, and successful ICMP ping tests in both directions between App-A and App-B.

## Key learnings.

" VPCs provide logically isolated IP networks in AWS.

" Subnets divide a VPC into smaller network segments.

" A bastion host can act as a controlled SSH entry point to private instances.

" A VPC peering connection provides a private path between two VPCs, but routes must be configured on both sides.

" Security groups control which inbound traffic is permitted; routes alone do not grant access.

" Connectivity tests before and after configuration help demonstrate the effect of the networking changes.

## Evidence not included in the supplied screenshots

The screenshots provided do not show the security-group rule-removal/restoration demonstration or final AWS cleanup. Those activities are therefore not claimed as completed in this report. Add evidence if you perform them and your workbook requires them.

# 8. Cleanup Reminder

After saving all evidence and when the lab is no longer needed, remove lab resources according to the workbook's cleanup guidance to avoid unnecessary AWS charges. Confirm dependencies before deleting network resources, and do not delete unrelated default VPC resources.